



PlayWise

Software Dependability

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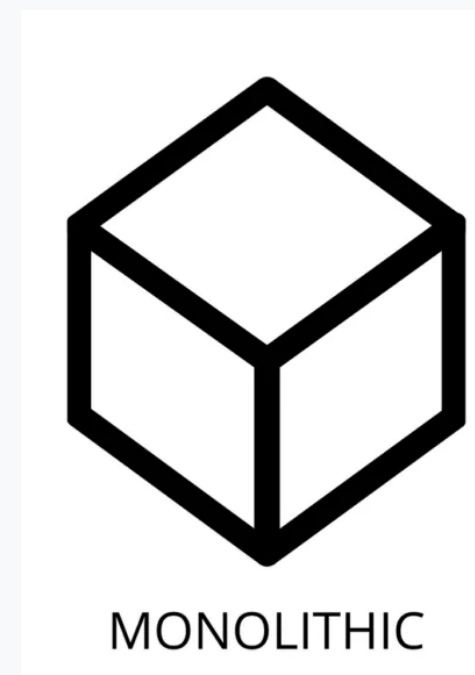
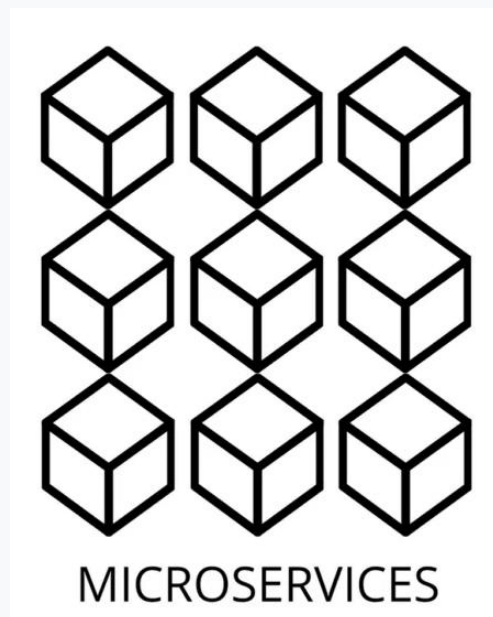
PlayWise Overview

PlayWise is a platform designed to:

- Offer a wide **catalog of video games** and an active user community
- Allow people to **share impressions** with other users on the platform



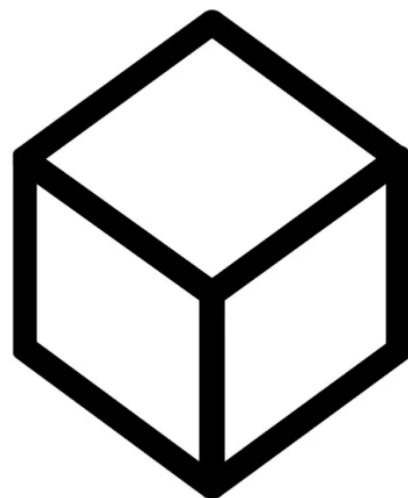
PlayWise Transformation



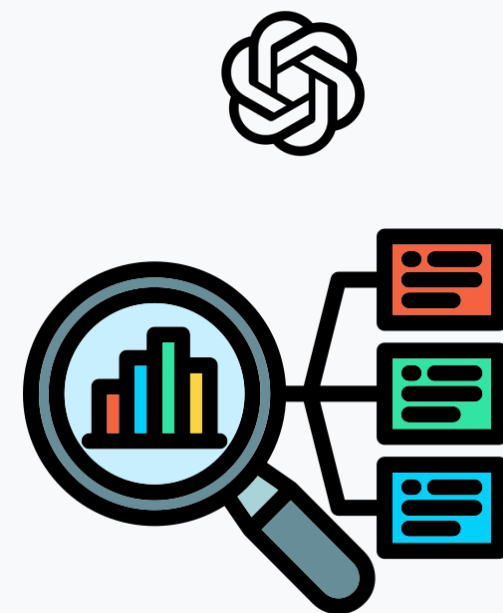
PlayWise Technologies



Overview



MONOLITHIC



PEGI classifier

Project's goals

With the application of **software dependability** concepts, we want to introduce:

- **Reliability**
- **Security**
- **Performance Reliability**
- **Availability**



Reliability



To ensure system **correctness** and **stability**.



Formal Verification

OpenJML

Static (**ESC**) and dynamic (**RAC**) verification successfully completed on the critical **GeneticEngine** module.



Code Coverage (Test with ChatGPT)

100%

Reached 100% Line and Branch coverage with **JaCoCo** on Controller, Service, and Utility modules.



Mutation testing

100%

Zero surviving mutants with **PITest**.
A robust test suite against logical faults.



To **detect vulnerabilities** and **dependency risks**.



Secret detection

GitGuardian

Detects exposed **secrets** and **credentials** in the repository, preventing accidental .



Vulnerability scan

Snyk

Scans project dependencies and configurations to identify and fix known security **vulnerabilities**.



Code quality

SonarCloud

Performs static code analysis to detect bugs, code smells, and security hotspots, improving **overall code quality**.



Dependency updates

Dependabot

Automatically monitors and updates vulnerable or outdated **dependencies** to maintain a secure software supply chain.

Performance Reliability



To measure **efficiency** and **performance stability**.



Test analytics

CodeCov

Collects coverage and test execution **metrics** to identify untested code **paths** and performance-critical tests.



Microbenchmarking

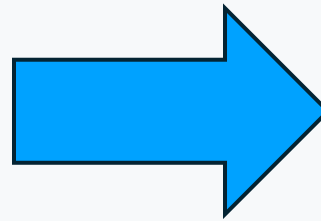
JMH

Measures **execution time** and **performance** of critical methods in isolation, enabling targeted optimizations.

Performance Reliability



Benchmark	Tempo medio (ms/op)	Errore
GameService		
GetGameDetails (5+ reviews)	1.715	1.598
ReviewController		
GetReviewsForGame	1.322	1.843
PageRequestCreation	$\approx 10^{-4}$	–
SortParsing	$\approx 10^{-4}$	–
UserService		
Login	74.446	0.495
PasswordEncode	74.431	0.463
PasswordMatch	74.071	0.461
Register	74.045	0.234
UpdateUser	0.019	0.003



Benchmark	Tempo medio (ms/op)	Errore
GameService		
GetGameDetails (5+ reviews)	1.740	2.110
ReviewController		
GetReviewsForGame	1.417	1.986
PageRequestCreation	$\approx 10^{-4}$	–
SortParsing	$\approx 10^{-4}$	–
UserService		
Login	1.203	0.002
PasswordEncode	1.198	0.003
PasswordMatch	1.198	0.006
Register	1.231	0.089
UpdateUser	0.019	0.002

Availability



To ensure continuous and automated **service operation**.



Dockerization

DockerHub

Published on **DockerHub** to
guarantee **reproducible**
deployment and high availability.

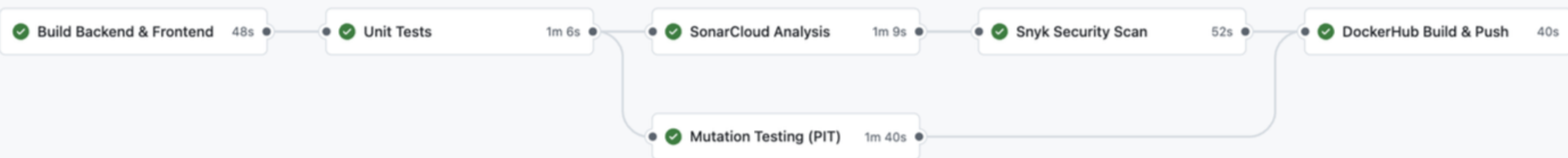


CI/CD

**GitHub
Actions**

CI/CD automates build, testing, and
deployment to ensure continuous
availability and reliability.

Availability





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